

SIGNED DIVISOR PAIRINGS, CROSS-SHELL BOUNDARY CLASSES, AND THE UNIQUE THREE-COLOUR RETURN FOR ERDŐS–STRAUS

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ABSTRACT. For every original Erdős–Straus shell we place the complete divisor source in its signed opposition pairing before selecting an integer witness. Its exact fibres are the original exterior and middle words with strictly positive, explicit divisor weights, so strict positivity is equivalent to shell occupancy. The character sign and the even–odd energy difference show why an ordinary receiver norm cannot replace this pairing. An exact cross-shell map produces a four-term reciprocal identity; its remaining term is classified by a finite cyclic obstruction whose order is the retained numerator of the failed unit-fraction fusion. We then prove a general complement law on an odd residual quotient. It decomposes the trace source into middle, exterior, and uncovered orbits and maps exactly to the earlier affine channel coordinate. Quotient three is the unique nontrivial odd quotient on which complementation alone forces every colour orbit into an integer gate. This gives a same-shell return theorem, an exact solution count, the complete applicable-residual family, and a factor-level classification of the empty residual-27 locus. The missing global step is now explicit: for each prescribed hard prime one must force a nonempty three-colour trace source or eliminate a cross-shell obstruction.

1. ORIGINAL SHELLS AND MARKED WORDS

Fix a prime

$$p \equiv 1 \pmod{24}$$

and an integer a with $p/4 < a < p/2$. Retain

$$R = 4a - p, \quad S = pa.$$

Then $0 < R < p$, $R \equiv 3 \pmod{4}$, and

$$\gcd(p, R) = \gcd(a, R) = \gcd(S, R) = 1. \quad (1.1)$$

The last equalities follow from $R = 4a - p$, $a < p$, and the primality of p . Every divisor of S therefore represents a unit modulo R .

For $u \mid a^2$, put

$$g_u = \gcd(a, u), \quad (h_u, r_u, s_u) = \left(\frac{g_u^2}{u}, \frac{u}{g_u}, \frac{a}{g_u} \right). \quad (1.2)$$

These are positive integers and retain the exact inverse data

$$a = h_u r_u s_u, \quad u = h_u r_u^2, \quad \gcd(r_u, s_u) = 1. \quad (1.3)$$

The original full word sets are

$$E_a = \{u \mid a^2 : R \mid 4u + 1\}, \quad M_a = \{u \mid a^2 : R \mid p + 4u\}. \quad (1.4)$$

Their marked ordered returns are

$$E_a : \quad \kappa = \frac{pr + s}{R}, \quad (x, y, z) = (a, hsk\kappa, phr\kappa), \quad (1.5)$$

$$M_a : \quad \lambda = \frac{r + s}{R}, \quad (x, y, z) = (a, phs\lambda, phr\lambda). \quad (1.6)$$

The classical type-I/type-II parametrizations are the literature antecedent (Elsholtz and Tao, 2013, Section 2, Propositions 2.2 and 2.6). Equations (1.5)–(1.6) retain the marked coordinates used in this work; no existence statement is imported from the parametrization.

2. THE COMPLETE ORIGINAL-SOURCE SIGNED PAIRING

Let $G_R = (\mathbb{Z}/R\mathbb{Z})^\times$. In the real Hilbert space with orthonormal basis $\{e_t : t \in G_R\}$, define, for every real σ ,

$$v_{p,a,\sigma} = \sum_{d|S} \left(\frac{d}{\sqrt{S}} \right)^{-\sigma} e_{d \bmod R}, \quad J_R e_t = e_{-t}. \quad (2.1)$$

Different divisors with the same residue remain as separate summands before their coefficients are added at the same basis vector. The vector exists at every shell, whether or not either gate in (1.4) is occupied. Set

$$\mathcal{T}_{p,a}(\sigma) = \langle v_{p,a,\sigma}, J_R v_{p,a,\sigma} \rangle = \sum_{\substack{d,e|pa \\ R|d+e}} \left(\frac{de}{pa} \right)^{-\sigma}. \quad (2.2)$$

The unweighted pair statistic and its character form occur in the earlier Boolean-product collaboration (JT and The Clankers, 2026, BP15–BP16); the next theorem computes all weighted fibres over the original marked words.

Theorem 2.1 (Exact positive-weight fibres). *For*

$$W_\sigma(h) = \sum_{k|h} \left(\frac{h}{k^2} \right)^\sigma,$$

one has, for every real σ ,

$$\boxed{\mathcal{T}_{p,a}(\sigma) = 2 \sum_{u \in E_a} W_\sigma(h_u) + (p^\sigma + p^{-\sigma}) \sum_{u \in M_a} W_\sigma(h_u).} \quad (2.3)$$

Consequently

$$\boxed{\mathcal{T}_{p,a}(\sigma) > 0 \iff E_a \cup M_a \neq \emptyset.} \quad (2.4)$$

At $\sigma = 0$ this becomes

$$\mathcal{T}_{p,a}(0) = 2 \sum_{u \in E_a} \tau_{\text{div}}(h_u) + 2 \sum_{u \in M_a} \tau_{\text{div}}(h_u). \quad (2.5)$$

Proof. Because $p \nmid a$, every divisor of pa is uniquely $p^\epsilon b$ with $\epsilon \in \{0, 1\}$ and $b \mid a$. For an ordered pair $b, c \mid a$, retain

$$k = \gcd(b, c), \quad b = kr, \quad c = ks, \quad \gcd(r, s) = 1.$$

The least common multiple krs divides a . Writing $h = a/(rs)$, the exact remaining condition is $k \mid h$. Conversely every $k \mid h$ returns the actual divisors $kr, ks \mid hrs = a$. The associated original marked divisor is

$$u = hr^2, \quad a = hrs. \quad (2.6)$$

The patterns with neither or both entries divisible by p are

$$(kr, ks), \quad (pkr, pks), \quad k \mid h.$$

Because k is a unit modulo R , their opposition congruence is $R \mid r + s$. Using $p = 4hrs - R$ and the fact that $4hr$ is a unit modulo R , this is equivalent to $R \mid p + 4hr^2$, the middle gate. The two retained weights are

$$p^\sigma \left(\frac{h}{k^2} \right)^\sigma, \quad p^{-\sigma} \left(\frac{h}{k^2} \right)^\sigma. \quad (2.7)$$

The mixed patterns are

$$(pkr, ks), \quad (ks, pkr), \quad k \mid h.$$

Here r is always the coprime part carried by the p -bearing divisor; the second displayed pair retains the opposite position bit. Conversely, for any mixed ordered pair, remove the one factor p , orient r from that entry and s from the other entry, and then recover k, h, u . This gives both position bits without omitting or duplicating an ordered pair. Their opposition congruence is $R \mid pr + s$. Since

$$pr + s = s(4hr^2 + 1) - Rr$$

and s is a unit modulo R , this is equivalent to the exterior gate $R \mid 4u + 1$. Each retained weight is

$$\left(\frac{h}{k^2}\right)^\sigma. \quad (2.8)$$

The four p -patterns exhaust every ordered pair in (2.2), proving (2.3). Every summand is strictly positive for real σ , which proves (2.4) and (2.5). $\square$

Proposition 2.2 (Character sign, parity energies, and degree action). *For a character $\chi \in \widehat{G}_R$, retain*

$$Z_{S,\chi}(\sigma) = \sum_{d \mid S} \chi(d) \left(\frac{d}{\sqrt{S}}\right)^{-\sigma}.$$

Then

$$\boxed{\mathcal{T}_{p,a}(\sigma) = \frac{1}{\varphi(R)} \sum_{\chi \in \widehat{G}_R} \chi(-1) |Z_{S,\chi}(\sigma)|^2.} \quad (2.9)$$

With $P_\pm = (I \pm J_R)/2$,

$$\boxed{\mathcal{T}_{p,a}(\sigma) = \|P_+ v_{p,a,\sigma}\|^2 - \|P_- v_{p,a,\sigma}\|^2.} \quad (2.10)$$

The map $e_t \mapsto e_{t/R}$ embeds this shell space into the finitely supported part of $\ell^2(\mathbb{Q}/\mathbb{Z})$. Its support consists of points of exact additive order R , so supports belonging to distinct residuals are disjoint. For

$$V_{p,\sigma} = \sum_{p/4 < a < p/2} v_{p,a,\sigma}$$

one therefore has

$$\langle V_{p,\sigma}, JV_{p,\sigma} \rangle = \sum_{p/4 < a < p/2} \mathcal{T}_{p,a}(\sigma). \quad (2.11)$$

On finitely supported functions define $(A_q f)(x) = f(qx)$. Then

$$\boxed{A_q^* J A_q = qJ.} \quad (2.12)$$

In particular the degree action preserves signed isotropy.

Proof. Expand $|Z_{S,\chi}|^2$ with the real positive divisor weights. Character orthogonality applied to $-de^{-1} \in G_R$ retains exactly $d + e \equiv 0 \pmod{R}$, which proves (2.9). The self-adjoint involution J_R has spectral projections $P_\pm$, giving (2.10). Since R is odd, both $e_t + e_{-t}$ and $e_t - e_{-t}$ occur, so the signed form is indefinite; it cannot be replaced on the full space by the pullback of a positive semidefinite form.

For $t \in G_R$, the class t/R has exact order R . This proves the disjoint support assertion and (2.11). Finally, each $y \in \mathbb{Q}/\mathbb{Z}$ has exactly q preimages under multiplication by q . Hence $\langle A_q f, A_q g \rangle = q \langle f, g \rangle$, while $A_q J = J A_q$. Equation (2.12) follows. $\square$

The operator in (2.12) is the unscaled global pullback; explicitly,

$$(A_q^* g)(y) = \sum_{qx=y} g(x), \quad A_q^* A_q = qI. \quad (2.13)$$

The scaled operator $V_q = q^{-1/2} A_q$ instead satisfies $V_q^* J V_q = J$. The global operator need not preserve one exact-order stratum: its compression to the order- R stratum is the permutation $t \mapsto qt$ when $\gcd(q, R) = 1$, and is zero when $\gcd(q, R) > 1$; all uncompressed support remains in its actual exact-order strata.

Equations (2.9)–(2.12) locate the sign that any analytic comparison must retain. A lower bound for the unsigned energy does not imply a strict lower bound for (2.10).

3. CROSS-SHELL BOUNDARY AND ITS OBSTRUCTION SPACE

Choose an integer $m \geq 1$, put $q = 4m + 1$, and assume $qR < p$. Retain two original shells

$$a = \frac{p+R}{4}, \quad a' = \frac{p+qR}{4} = a + mR. \quad (3.1)$$

Take actual positive divisors

$$d \mid pa, \quad e \mid pa', \quad R \mid d + e, \quad (3.2)$$

and their actual complements

$$\alpha = \frac{pa}{d}, \quad \beta = \frac{pa'}{e}. \quad (3.3)$$

By (1.1), d is a unit modulo R . Since $e\beta - d\alpha = pa' - pa = pmR$ and $e \equiv -d \pmod{R}$, cancellation of d gives $R \mid \alpha + \beta$. Thus the following are positive integers:

$$\lambda = \frac{d+e}{R}, \quad \mu = \frac{\alpha+\beta}{R}. \quad (3.4)$$

Theorem 3.1 (Exact cross-shell boundary). *The retained data satisfy both determinant identities*

$$\boxed{\lambda\beta - d\mu = pm, \quad e\mu - \lambda\alpha = pm} \quad (3.5)$$

and the four-term reciprocal identity

$$\boxed{\frac{4}{p} = \frac{1}{a} + \frac{1}{\lambda\alpha} + \frac{1}{\mu d} + \frac{m}{a\lambda\mu}}. \quad (3.6)$$

Let

$$L = \lambda\mu, \quad g = \gcd(L, m), \quad \ell = \frac{L}{g}, \quad \nu = \frac{L+m}{g}. \quad (3.7)$$

Then $\gcd(\ell, \nu) = 1$, and the first and fourth terms in (3.6) form one unit fraction exactly when

$$\boxed{\nu \mid a}. \quad (3.8)$$

On this exact domain,

$$\boxed{X = \frac{a\lambda\mu}{\lambda\mu + m} = \frac{a}{\nu} \frac{\lambda\mu}{g}, \quad Y = \lambda\alpha, \quad Z = \mu d} \quad (3.9)$$

are positive integers and $4/p = 1/X + 1/Y + 1/Z$.

Proof. The first identity in (3.5) is

$$\lambda\beta - d\mu = \frac{(d+e)\beta - d(\alpha+\beta)}{R} = \frac{e\beta - d\alpha}{R} = pm.$$

The second is obtained by retaining the other column. Since $d\alpha = pa$,

$$\frac{1}{\lambda\alpha} + \frac{1}{\mu d} + \frac{m}{a\lambda\mu} = \frac{\mu d + \lambda\alpha + pm}{pa\lambda\mu} = \frac{\lambda(\alpha+\beta)}{pa\lambda\mu} = \frac{R}{pa}.$$

Adding $1/a$ and retaining $p+R=4a$ proves (3.6).

The retained fusion is

$$\frac{1}{a} + \frac{m}{aL} = \frac{L+m}{aL} = \frac{\nu}{a\ell}.$$

Because $\gcd(\nu, \ell) = 1$, its reciprocal $a\ell/\nu$ is integral exactly when $\nu \mid a$. Equations (3.9) then retain all three denominators. $\square$

Definition 3.2 (Boundary obstruction). *For fixed m, λ, μ , let*

$$\mathcal{O}_\nu = (\nu^{-1}\mathbb{Z})/\mathbb{Z}, \quad \Theta_{\ell, \nu} : \mathbb{Z} \longrightarrow \mathcal{O}_\nu, \quad n \longmapsto \frac{n\ell}{\nu} + \mathbb{Z}. \quad (3.10)$$

The exact sequence

$$0 \longrightarrow \nu\mathbb{Z} \longrightarrow \mathbb{Z} \xrightarrow{\Theta_{\ell, \nu}} \mathcal{O}_\nu \longrightarrow 0 \quad (3.11)$$

defines the cyclic obstruction class

$$\Omega(a; m, \lambda, \mu) = \Theta_{\ell, \nu}(a) = \frac{a\lambda\mu}{\lambda\mu + m} + \mathbb{Z}. \quad (3.12)$$

It vanishes exactly on the fusion domain (3.8).

Put $h = \gcd(a, \nu)$, $A = a/h$, and $\rho = \nu/h$. The complete retained defect is

$$\frac{1}{a} + \frac{m}{aL} = \frac{\rho}{A\ell}, \quad \gcd(\rho, A\ell) = 1, \quad \text{ord } \Omega = \rho. \quad (3.13)$$

Thus a failed fusion is not discarded: its exact reduced numerator is the order of a point in $\mathcal{O}_\nu$. The order- ρ stratum is

$$\mathcal{S}_\rho = \left\{ \frac{\nu}{\rho} w \pmod{\nu} : w \in (\mathbb{Z}/\rho\mathbb{Z})^\times \right\}, \quad (3.14)$$

with $\varphi(\rho)$ residue classes. In the original shell variables,

$$\nu \mid a \iff 4\nu \mid p + R, \quad \rho = \frac{4\nu}{\gcd(p + R, 4\nu)}. \quad (3.15)$$

The surviving identity outside the fusion locus remains

$$\frac{4}{p} = \frac{1}{\lambda\alpha} + \frac{1}{\mu d} + \frac{\rho}{A\ell}. \quad (3.16)$$

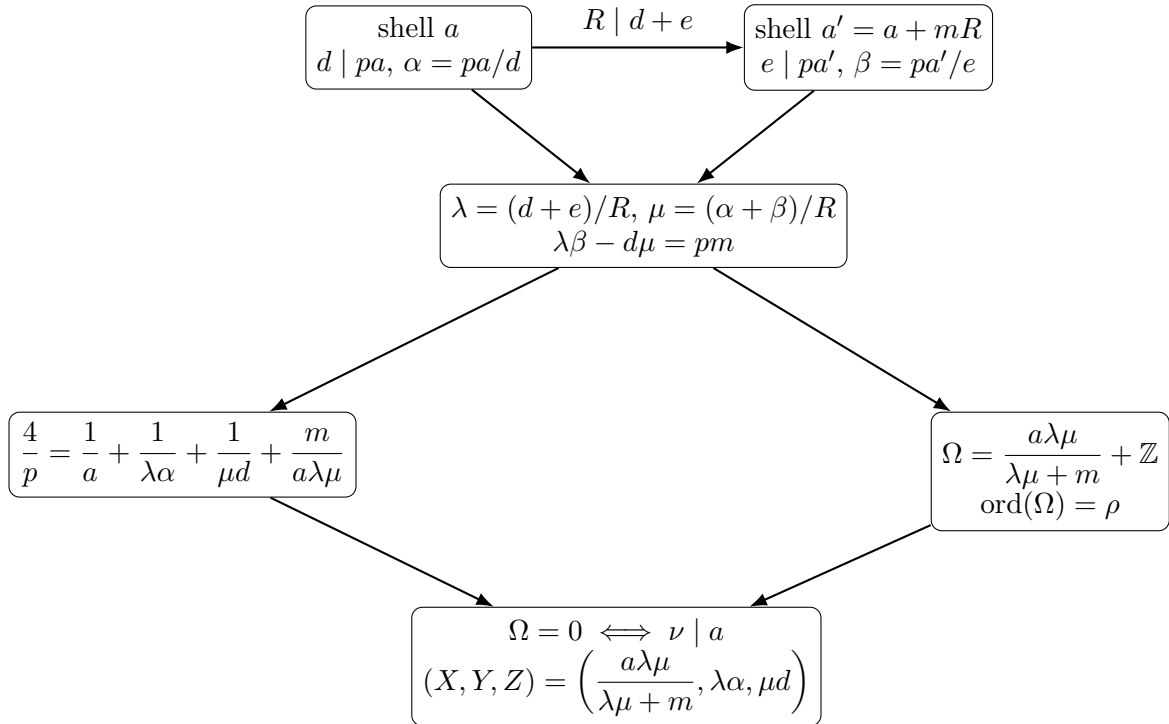


FIGURE 1. Every displayed arrow retains its domain and codomain. The cross-shell pair produces the determinant pm and the fourth reciprocal term. Fusion is an exact map only on the kernel of $\Theta_{\ell, \nu}$; outside it, the order ρ records the surviving boundary numerator.

4. GENERAL ODD-QUOTIENT COMPLEMENT LAW

The same-shell theorem is best understood before setting the quotient equal to three. Continue with one original shell (p, a, R) . Suppose a positive divisor $m \mid R$ satisfies

$$\gcd(R, p - 1) = m, \quad R \mid m^2, \quad N = R/m > 1. \quad (4.1)$$

All of R, m, N are odd. Define the complete trace source

$$\mathcal{T}_N = \{u \mid a^2 : m \mid p + 4u\}, \quad (4.2)$$

the complement $\iota(u) = a^2/u$, and

$$b_u = \frac{p + 4u}{m} \pmod{N}, \quad \omega = \frac{p - 1}{m} \pmod{N}. \quad (4.3)$$

Theorem 4.1 (Exact N -colour complement decomposition). *Under (4.1), ω is a unit in $\mathbb{Z}/N\mathbb{Z}$, ι is a fixed-point-free involution of $\mathcal{T}_N$, and*

$$\boxed{b_{\iota(u)} = -b_u \pmod{N}.} \quad (4.4)$$

The full gates are

$$u \in M_a \iff b_u = 0, \quad u \in E_a \iff b_u = \omega. \quad (4.5)$$

Let

$$U_N = \{u \in \mathcal{T}_N : b_u \notin \{0, \omega, -\omega\}\}.$$

Then U_N is complement-stable, and there is an exact labelled orbit decomposition

$$\boxed{\mathcal{T}_N/\iota \simeq E_a \amalg (M_a/\iota) \amalg (U_N/\iota).} \quad (4.6)$$

Consequently

$$\boxed{|\mathcal{T}_N| = 2|E_a| + |M_a| + |U_N|.} \quad (4.7)$$

The inverse in (4.6) attaches the actual pair $\{u, a^2/u\}$; an input-member bit restores the labelled source word.

Proof. The exact gcd in (4.1) gives

$$\gcd\left(N, \frac{p-1}{m}\right) = 1,$$

so ω is a unit. For $u \in \mathcal{T}_N$,

$$4(a + u) = p + R + 4u$$

is divisible by m . Since m is odd, $m \mid a + u$; since $R \mid m^2$, $R \mid (a + u)^2$. The divisor u is a unit modulo R , and

$$u + \frac{a^2}{u} + 2a = \frac{(a + u)^2}{u} \equiv 0 \pmod{R}. \quad (4.8)$$

Therefore

$$(p + 4u) + \left(p + 4\frac{a^2}{u}\right) \equiv 0 \pmod{R},$$

which proves that $\iota(u) \in \mathcal{T}_N$ and gives (4.4) after division by m .

A fixed point would have $u = a$. Membership in $\mathcal{T}_N$ would then give $m \mid p + 4a = 2p + R$, hence $m \mid 2p$. But $m > 1$ is odd and is coprime to p , a contradiction. Finally,

$$p + 4u - (p - 1) = 4u + 1$$

proves (4.5). Every orbit with colour zero consists of two middle words; every orbit with colours $\{\omega, -\omega\}$ contains exactly one exterior word; the remaining orbits are exactly U_N/ι . This proves (4.6)–(4.7). $\square$

Corollary 4.2 (Uniqueness of three colours). *Complementation alone guarantees that every possible colour orbit meets a full E or M gate if and only if $N = 3$. More exactly,*

$$\{0, \omega, -\omega\} = \mathbb{Z}/N\mathbb{Z} \iff N = 3. \quad (4.9)$$

For a particular source at $N > 3$, return still holds exactly when its realized colour support is contained in $\{0, \omega, -\omega\}$; the nonempty set U_N is the exact obstruction otherwise.

Proof. The three displayed residues are distinct because $N > 1$ is odd and ω is a unit. Equality with the entire residue group therefore forces $N = 3$. Conversely a group of order three consists exactly of those residues. $\square$

Proposition 4.3 (Sharpness for every larger odd quotient). *For every odd $N > 3$, there are infinitely many primes $p \equiv 1 \pmod{24}$ and original shells satisfying (4.1) whose source $\mathcal{T}_N$ contains a wholly uncovered complement orbit.*

Proof. Set

$$m = 3N, \quad R = 3N^2 = mN. \quad (4.14)$$

Then $R \equiv 3 \pmod{4}$ and $R \mid m^2$. By the Chinese remainder theorem, choose $s > 0$ with

$$s \equiv 1 \pmod{N}, \quad ms \equiv 5 \pmod{8}, \quad (4.15)$$

and retain

$$U = \frac{ms - 1}{4}. \quad (4.16)$$

Thus U is odd and $\gcd(U, mN) = 1$. Choose an even k_0 with

$$4Uk_0 \equiv 2 \pmod{N}, \quad (4.17)$$

which is possible because $4U$ is a unit modulo odd N and adding N changes parity. For $k = k_0 + 2Nj$, define

$$a = U(mk - 1), \quad p = 4a - R. \quad (4.18)$$

As j varies, p lies in the progression

$$p = p_0 + 8UmNj. \quad (4.19)$$

The retained identities give $p \equiv 1 \pmod{8}$ and $p \equiv 1 \pmod{3}$, hence $p \equiv 1 \pmod{24}$. Moreover $p \equiv 1 \pmod{m}$, $p \equiv -R \pmod{U}$, and $\gcd(U, R) = 1$, so

$$\gcd(p_0, 8UmN) = 1.$$

Dirichlet's theorem on primes in reduced progressions (Sutherland, 2017, Theorem 18.1) supplies infinitely many prime values; choose one with $p > R$.

Using $4U = ms - 1$, one has

$$p - 1 = m(4Uk - s - N), \quad \frac{p - 1}{m} \equiv 1 \pmod{N}. \quad (4.20)$$

Therefore $\gcd(R, p - 1) = m$ and $\omega = 1$. The actual divisor $U \mid a^2$ satisfies

$$p + 4U = m(4Uk - N), \quad b_U = 2 \pmod{N}. \quad (4.21)$$

Its complement has colour -2 . For odd $N > 3$, neither colour is 0 or $\omega = 1$, so this entire complement orbit lies in U_N . $\square$

Remark 4.4 (The quotient-one edge). *If $N = 1$, then $m = R$ and $\mathcal{T}_1 = M_a = E_a$: the two gates coincide because $R \mid p - 1$. In the original shell domain $R \equiv 3 \pmod{4}$, hence $R > 1$, and the divisor complement is free. This doubly gated edge is separate from the nontrivial odd-quotient uniqueness in Corollary 4.2.*

4.1. Exact morphism to the affine channel coordinate. Let

$$K = \prod_{\ell^e \parallel R} \ell^{\lceil e/2 \rceil}, \quad D = R/K. \quad (4.10)$$

The hypotheses $R = mN$ and $R \mid m^2$ first give $N \mid m$. They also imply, prime by prime,

$$K \mid m, \quad N \mid K. \quad (4.11)$$

Write $t = m/K$; then $D = Nt$. The source $\mathcal{T}_N$ lies in the common E/M trace locus of the affine-channel theorem because $K \mid m$ divides both $p - 1$ and $p + 4u$. Its middle fine coordinate is

$$c_M \equiv p^{-1} \frac{p + 4u}{K} \pmod{D}.$$

Since $D \mid m$ and $m \mid p - 1$, one has $p^{-1} \equiv 1 \pmod{D}$. Thus

$$c_M = t b_u \pmod{Nt}. \quad (4.12)$$

The exact subgroup quotient

$$t\mathbb{Z}/(Nt)\mathbb{Z} \longrightarrow \mathbb{Z}/N\mathbb{Z}, \quad tb \longmapsto b \quad (4.13)$$

sends the prior affine E target $(p-1)/K = t\omega$ to ω , the M target to zero, and the divisor complement to $b \mapsto -b$. This is the exact morphism between the fine-defect channel operator and the colour-orbit theorem; neither presentation replaces the other.

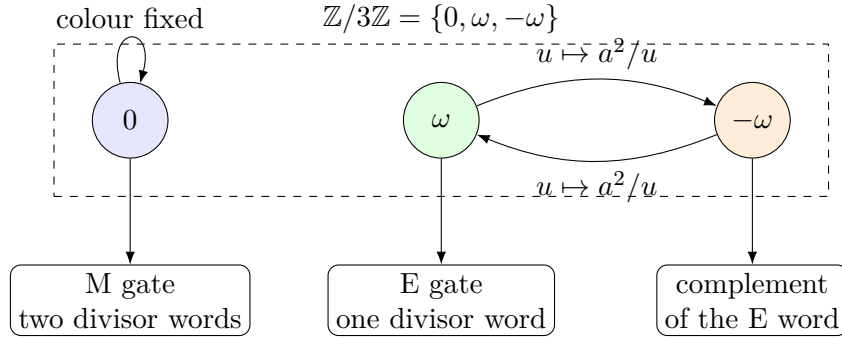


FIGURE 2. The exact quotient-three mechanism. At residue level, complementation fixes colour zero and exchanges ω with $-\omega$; on divisor words it is fixed-point-free. The zero orbit contains two M words, while the nonzero orbit contains exactly one E word. Larger odd quotients have additional uncovered orbits.

5. SAME-SHELL THIRD-DEFECT RETURN

Assume now

$$\boxed{\gcd(R, p-1) = R/3}, \quad m = R/3. \quad (5.1)$$

Because $p \equiv 1 \pmod{3}$, valuation comparison forces $9 \mid R$; hence $3 \mid m$ and $R = 3m \mid m^2$. Thus Theorem 4.1 applies with $N = 3$ to the complete source

$$\mathcal{T}_3 = \{u \mid a^2 : m \mid p + 4u\}. \quad (5.2)$$

Theorem 5.1 (Whole-shell third-defect completion). *Every member of $\mathcal{T}_3$ gives a rational middle trace whose two tails have denominators dividing three. It returns at unchanged (p, R, a) to an original integer E or M word using either u or a^2/u . Explicitly, with*

$$b_u = \frac{p+4u}{m} \pmod{3}, \quad \omega = \frac{p-1}{m} \pmod{3},$$

the return is

b_u	marked integer output
0	M at u
ω	E at u
$-\omega$	E at a^2/u .

(5.3)

Moreover

$$\boxed{2|E_a| + |M_a| = |\mathcal{T}_3|}, \quad (5.4)$$

and therefore

$$\boxed{E_a \cup M_a = \emptyset \iff \mathcal{T}_3 = \emptyset}. \quad (5.5)$$

The number of increasing original denominator triples at this shell is exactly $|\mathcal{T}_3|/2$.

Proof. Only the tail and increasing-order assertions remain beyond Theorem 4.1 and Corollary 4.2. Put $v = a^2/u$. The rational middle tails are retained as

$$y = \frac{p(a+v)}{R}, \quad z = \frac{p(a+u)}{R}. \quad (5.6)$$

Both u and v belong to $\mathcal{T}_3$, so m divides both $a+u$ and $a+v$; since $R = 3m$, each tail has reduced denominator dividing three. Their sum retains the exact integer formula

$$y+z = \frac{p(a+u)^2}{Ru}. \quad (5.7)$$

Indeed, $R \mid (a+u)^2$, $u \mid (a+u)^2$, and $\gcd(R, u) = 1$.

For an exterior word, (1.5) gives $a < y < z$: the inequalities are equivalent to $\kappa > r$ and $pr > s$, which follow from $p > R$ and $s < a < p$. Exactly one tail is divisible by p : the factor p is explicit in z , whereas $R\kappa = pr + s$ and $0 < s < p$ show that $p \nmid \kappa$, hence $p \nmid y$. The ordered inverse recovers its unique exterior word. For a middle word, both tails in (1.6) exceed a and are divisible by p ; complementation swaps r, s and hence swaps the two tails. There is no fixed middle word in $\mathcal{T}_3$. Therefore exterior words count once and middle complement pairs count once:

$$\#\{\text{increasing triples}\} = |E_a| + \frac{|M_a|}{2} = \frac{|\mathcal{T}_3|}{2}.$$

□

Under complementation, the primitive coordinates change exactly by

$$(h, r, s) \mapsto (h, s, r). \quad (5.8)$$

The prime, residual, distinguished denominator, and original exponent box are unchanged; the E/M quotient is recomputed in its own channel.

6. ALL APPLICABLE RESIDUALS FROM THE PRESCRIBED PRIME

Write the complete factorization

$$p - 1 = 2^k 3^\eta b, \quad \gcd(b, 6) = 1, \quad k \geq 3, \quad \eta \geq 1. \quad (6.1)$$

Theorem 6.1 (Applicable residual classification). *The positive residuals $R < p$, $R \equiv 3 \pmod{4}$ satisfying $\gcd(R, p - 1) = R/3$ are exactly*

$$\boxed{R = 3^{\eta+1}d, \quad d \mid b, \quad d \equiv (-1)^\eta \pmod{4}.} \quad (6.2)$$

Every one satisfies

$$R \leq \frac{3(p-1)}{2^k} < p. \quad (6.3)$$

The family is nonempty exactly when η is even or b has a prime divisor congruent to 3 modulo 4.

Proof. At the prime three, the equality

$$\min(v_3(R), \eta) = v_3(R) - 1$$

forces $v_3(R) = \eta + 1$. At every other odd prime, its exponent in R is at most its exponent in $p - 1$. This yields $R = 3^{\eta+1}d$ with $d \mid b$. The retained shell congruence $R \equiv 3 \pmod{4}$ is exactly $d \equiv (-1)^\eta \pmod{4}$. These conditions also prove the converse gcd identity. Since $d \leq b$, (6.3) follows. If η is even, $d = 1$ works. If η is odd, a divisor $d \equiv 3 \pmod{4}$ exists exactly when one of the prime divisors of b is 3 modulo 4. □

This theorem constructs every candidate residual from the prescribed prime. It does not assert that one of the associated sets $\mathcal{T}_3$ is nonempty.

7. COMPLETE FACTOR-LEVEL FAILURE LOCUS AT RESIDUAL 27

Theorem 7.1 (Residual-27 classification). *Let p be prime with*

$$\boxed{p \equiv 73 \text{ or } 145 \pmod{216}}, \quad a = \frac{p + 27}{4}. \quad (7.1)$$

The complete original $R = 27$ shell is empty if and only if one of the following disjoint factorization types occurs:

- (1) every prime factor of a is 1 modulo 3;
- (2) exactly two prime-factor occurrences of a are 5 modulo 9, counting multiplicity, and every remaining prime factor is 1 modulo 9.

Outside these two types, an original E or M word is constructed with at most three signed prime occurrences in u/a , followed by (5.3).

Proof. The two classes in (7.1) are exactly the classes $p \equiv 1 \pmod{24}$ for which $\gcd(27, p-1) = 9$. They give $a \equiv 7 \pmod{9}$ and

$$\mathcal{T}_3 = \{u \mid a^2 : u \equiv 2 \pmod{9}\}. \quad (7.2)$$

Write

$$a = \prod_q q^{e_q}, \quad u = \prod_q q^{e_q + \beta_q}, \quad -e_q \leq \beta_q \leq e_q. \quad (7.3)$$

The exact target is

$$\prod_q q^{\beta_q} \equiv -1 \pmod{9}. \quad (7.4)$$

The unit group modulo nine is generated by two; the residues

$$1, 2, 4, 8, 7, 5$$

have logarithms 0, 1, 2, 3, 4, 5 modulo six. The logarithm of a is four and the target in (7.4) is three.

A residue-8 prime supplies the target with one occurrence. Suppose none is present. Let H be the total multiplicity of primes in the residue classes 2, 5 modulo nine. The logarithm of a is even, so H is even. If $H \geq 4$, select three such occurrences, assigning sign +1 to residue two and sign -1 to residue five; each contributes logarithm one. If $H = 2$ and a residue 4 or 7 occurrence exists, one signed odd logarithm contributes one and one signed even logarithm contributes two.

If $H = 0$, every available logarithm is even, so the target three is impossible; this is the first exceptional type. In the remaining case $H = 2$, with no residue 8, 4, or 7, the total logarithm four forces both odd occurrences to be residue five. Their complete centred exponent range contributes only

$$\{-2, -1, 0, 1, 2\} \pmod{6},$$

which misses three. This is the second type. Every successful selection obeys the literal bounds in (7.3), so it returns an actual divisor of a^2 . $\square$

The exceptional examples

$$(p, a) = (1009, 259 = 7 \cdot 37), \quad (p, a) = (73, 25 = 5^2) \quad (7.5)$$

have, respectively, the first and second factorization type. Their residual-27 shells are empty; neither prime is an Erdős–Straus counterexample.

8. WHAT REMAINS

The results above are structural rather than a finite prime census. They give:

- (1) a complete source that exists at every shell and whose signed value is strictly positive exactly when that shell has an original integer word;
- (2) an exact cross-shell map with a classified cyclic boundary obstruction;
- (3) an exact same-shell quotient law in which three is the unique nontrivial odd quotient closed by the divisor complement alone;
- (4) every residual on which that three-colour theorem applies, constructed from the prescribed prime; and
- (5) the complete emptiness criterion for the first canonical residual 27.

They do not prove that an applicable $\mathcal{T}_3$ is nonempty for every hard prime, nor that a cross-shell pair with zero boundary obstruction must occur. An arbitrary counterexample would have to make every constructed three-colour source empty, defeat all other original shells, and avoid every cross-shell kernel in (3.11). Proving that this simultaneous avoidance is impossible is the remaining existence problem.

REPRODUCIBILITY AND ATTRIBUTION

The received four-page proof and constructor are preserved byte-for-byte under `received/`. The active constructor corrects two output-field names but does not alter the mathematics. A separately written checker reconstructs the complement law, count, residual family, residual-27 factor classification, signed pairing at fixed exact test values, and cross-shell identities without importing the active constructor. The Boolean-pair antecedent is credited to the JT collaboration; the type-I/type-II antecedent is credited to Elsholtz and Tao. No historical-priority claim is made for the new quotient and obstruction formulations.

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